

Technical Requirements Document (TRD)

Universal E-Reader Agent – Production Migration

Objective

Refactor the existing Kindle Agent into a production-ready, platform-independent document processing system.

Existing Architecture

React

↓

FastAPI

↓

Agent

↓

MCP

↓

OCR

↓

Calibre

↓

SMTP

↓

Kindle

Target Architecture

Browser



React



FastAPI



Agent



MCP



Document Processing



Delivery Adapter Layer



Platform

Delivery Adapter Layer

Introduce a new abstraction.

Interface:

DeliveryProvider

Responsibilities:

Accept generated EPUB

Deliver according to platform

Return delivery result

Initial Implementations

KindleDelivery

Implementation:

SMTP

↓

Kindle Email

DownloadDelivery

Implementation:

Return EPUB download

Future Implementations

KoboDelivery

- Dropbox integration (where available)
- USB export support (manual workflow)

PocketBookDelivery

- PocketBook Cloud integration (if available)
- Download fallback

AppleBooksDelivery

- Export EPUB
- User import workflow

Additional providers shall implement the same interface.

Agent Refactoring

Current:

Agent

↓

send_to_kindle()

Target:

Agent

↓

DeliveryProvider.deliver()

The agent must never reference Kindle-specific logic directly.

Configuration

Replace hardcoded values with environment variables.

Examples:

SMTP_HOST

SMTP_PORT

SMTP_USER

SMTP_PASSWORD

UPLOAD_LIMIT

TEMP_DIRECTORY

OUTPUT_DIRECTORY

FRONTEND_URL

Deployment

Frontend

- Vercel

Backend

- Dockerized FastAPI

Development

- Local

Production

- Free hosting platform or self-hosted
-

Storage

Current:

uploads/

outputs/

Retain for MVP.

Storage implementation shall remain abstract to allow future migration to cloud object storage.

Security

Validate uploads

Limit file size

Restrict MIME types

Sanitize filenames

Restrict CORS

Secure SMTP credentials

HTTPS only

Logging

Log:

Upload

Analysis

OCR

Conversion

Delivery

Failures

Testing

Verify:

PDF upload

OCR detection

OCR execution

Metadata extraction

EPUB generation

Delivery adapter execution

Kindle delivery

Download delivery

Health endpoint

Docker build

Environment loading

Future Expansion

The architecture shall support:

Additional document formats

Additional e-reader adapters

Background workers

Job queues

Database

Authentication

Cloud storage

without requiring modifications to the core agent orchestration logic.